

REFINERY PRESSURE VESSEL INSPECTION REPORT

Facility: Northern PSU Refinery Unit 4
Equipment Tag: PV-204B (Secondary Hydrocracker)
Inspection Standard: API 510 Pressure Vessel Inspection Code
Status: Scheduled Ultrasonic & Visual Wall Thickness Evaluation

1. Wall Thickness Ultrasonic Measurements

Inspection Zone	Nominal (mm)	Measured (mm)	Min Required (mm)	Compliance
Shell Ring 1	38.50	37.85	32.00	PASS
Shell Ring 2	38.50	36.90	32.00	PASS
Top Head Crown	42.00	41.10	35.50	PASS
Bottom Nozzle N1	25.40	22.80	21.00	MONITOR

2. Governing Formula

Minimum required thickness is calculated using ASME Section VIII Div 1 formula:
 $t_{min} = (P * R) / (S * E - 0.6 * P)$

3. Inspector Sign-off

Inspector ID: INSP-7749
Recommendation: Re-inspect nozzle N1 at 12-month interval. All other shell rings approved for service.